Satellite Communication Management System (SCMS)

Project Abstract — Database Management Systems

Problem Statement

Satellite operators, ground station networks and service providers depend on limited resources: orbital satellites, transponder bandwidth, frequency bands and short communication windows when a satellite passes over a ground station. Much of this is still managed with separate spreadsheets, old tools and manual scheduling, which leads to frequency conflicts, double-booked ground stations, wasted bandwidth and slow responses to anomalies. Each satellite also sends a constant stream of telemetry with varied structure (power, temperature, attitude, signal strength), which a single traditional database stores and analyzes poorly.

Motivation

Satellite communication supports navigation, weather forecasting, disaster response, broadband access and defense. With thousands of new satellites in orbit, managing them has become a real data-management challenge. The problem requires both strongly consistent transactional data (who owns which frequency, and when) and fast-growing, loosely structured data (telemetry and event logs), making it an ideal case for a hybrid database system that goes well beyond a basic CRUD application.

Proposed Solution

SCMS is a web-based platform that unifies satellite, ground station, scheduling and telemetry management. It follows a polyglot persistence architecture using two databases:

- **SQL database (MySQL / PostgreSQL)** — the system of record for structured, related data: satellites, orbits, transponders, frequency bands, ground stations, operators, clients, contracts, communication-window bookings and billing. The schema is normalized to 3NF/BCNF with referential integrity enforced by constraints. It uses stored procedures for bandwidth allocation, triggers that block overlapping frequency or time-slot bookings, views for role-based reports, indexing for fast schedule lookups, and ACID transactions with concurrency control so that two operators cannot reserve the same resource at once.
- **NoSQL database (MongoDB)** — stores high-volume, schema-flexible data: real-time telemetry packets, satellite health logs, anomaly reports and event histories. Its document model accommodates the different telemetry formats of different satellite models, while aggregation pipelines and time-based indexing support trend analysis and fast retrieval of recent health data.

Objectives

1. Design a normalized relational schema for managing satellite communication resources.
2. Prevent scheduling and frequency conflicts through database-level constraints, triggers and transactions.
3. Store and analyze heterogeneous telemetry data efficiently in a NoSQL database.
4. Integrate both databases into a single application with secure role-based access.
5. Provide operators with analytical insight into resource usage and satellite health.

Main Features

- Satellite and ground station registry with orbital and technical details.
- Communication-window scheduling with automatic conflict detection.
- Transponder and frequency-band allocation to clients, with contract and billing management.
- Telemetry ingestion and live satellite health monitoring.
- Threshold-based anomaly detection and alert logging.
- Role-based access for administrators, operators and clients.
- Dashboards and reports: bandwidth usage, ground station load, revenue per client and telemetry trends.

Expected Outcome

SCMS will replace scattered manual processes with a single reliable, data-driven system, resulting in fewer scheduling and frequency conflicts, better utilization of bandwidth and ground station time, faster detection of satellite anomalies, and a clear audit trail of all operations. Academically, the project demonstrates relational database design and normalization, advanced SQL (procedures, triggers, views, transactions and indexing), NoSQL data modeling and aggregation, and the integration of a polyglot persistence architecture into a working application.